Gaikwad Onkar Chandrakant

Bachelor of Technology in Artificial Intelligence | Indian Institute of Technology Gandhinagar | 2024–2028
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Education

Degree	Institution	CPI/%	Year
B.Tech	Indian Institute of Technology Gandhinagar	6.46	2024–2028

Experience

Flutter Developer – INSIIT **2025–2026**

Official Student Application, IIT Gandhinagar

- Contributed Flutter features to the official IIT Gandhinagar student app, serving 1,000+ active users, in a Git-based team workflow.
- Built Android home screen widgets using Flutter, native Kotlin/XML, and the `home_widget` package to display campus information.

Projects

Anar-X *Pomegranate Disease Detection using CNN* Live | GitHub

Python | PyTorch | OpenCV | Computer Vision | AGMARKNET

- Trained a **multi-task CNN** with 10.04M parameters on 5,099 pomegranate images, achieving 96.37% disease-classification accuracy with separate disease and severity prediction heads.
- Built a Python/OpenCV inference pipeline for image preprocessing, disease prediction, confidence scoring, and treatment recommendations.
- Integrated the AGMARKNET REST API to display mandi prices alongside disease diagnoses.

Aera Messenger *AI-Powered Messaging and Semantic Search* Live | GitHub

Flutter | Firebase | Supabase | Gemini API | Vector Embeddings

- Built a cross-platform Flutter messenger using Firebase Firestore for real-time messaging and Supabase for media storage and persistent user data.
- Implemented semantic search with vector embeddings to retrieve conversation messages by meaning instead of exact keywords.
- Separated Gemini API calls into a dedicated service layer to keep AI logic independent of the UI.

SmartSpend AI *AI-Powered Personal Expense Management* Live | GitHub

Next.js | TypeScript | Supabase | PostgreSQL | Gemini API | Zustand

- Used Supabase Auth and PostgreSQL Row-Level Security (RLS) to isolate each user's financial records at the database level.
- Built an offline-first PWA with Next.js, TypeScript, and Zustand, supporting local transaction entry and later synchronization.
- Integrated Gemini API to generate personalized budgeting recommendations and spending insights.

NEXO *LLM-Based Autonomous Task Automation* Live | GitHub

Python | FastAPI | Playwright | Next.js | LLM Agents | SQLite

- Built an LLM-driven agent that converts natural-language goals into multi-step plans, executes tasks, observes results, and adapts to errors, with a rule-based fallback.
- Automated a simulated company suite covering documents, finance, CRM, and email using Playwright, with error recovery and final-state verification.
- Added human-approval checkpoints for sensitive actions and streamed execution traces to a Next.js console using Server-Sent Events (SSE).

Technical Skills

Languages	Python, SQL, C++, C, JavaScript, Dart
Databases & Backend	PostgreSQL, SQLite, Supabase, Firebase, RLS, FastAPI, REST APIs
ML, CV & LLMs	Machine Learning, Deep Learning, Computer Vision, PyTorch, OpenCV, Vector Embeddings, LLM APIs
Frameworks & Libraries	Flutter, Next.js, React, Progressive Web Apps (PWA)
Tools & Fundamentals	Git/GitHub, Docker, Linux, Data Structures & Algorithms

Achievements

- **Astronomy** – Core Member, Odyssey Astronomy Club, IITGN
- **Vibe2Ship Hackathon** – Qualifier
- **Bhartiya Antariksh Hackathon 2026** – Team Lead

Extracurricular Activities

- **Sports** – Swimming and Esports
- **Photography** – Maintain an Instagram page to share my photographs